

Question	Answer	Additional guidance	Mark
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Number			
3(a)	A description that includes the following points: • modifies the {enzyme / protein / polypeptide} (1) • packages enzyme into a (secretory) vesicle (1)	ACCEPT enzyme(s) {leave (cell) / secreted} by exocytosis ACCEPT enzymes leave cell via secretory vesicle	(2)

Question Number	Answer	Additional guidance	Mark
3(b)	An answer that includes two of the following points: • for nucleic acid production (1) • amino acids for {enzyme / hormone / (structural) protein} production (1)	ignore nitrates e.g. to make (organic) {bases / nucleotides / DNA/ RNA / modified nucleotide e.g. ATP} accept amino acids {to form a named protein / for protein synthesis} accept to convert {glucose / product of photosynthesis} into another molecule (1) accept to form chlorophyll	(2)

Question	Answer	Mark
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Number		
3(c)(i)	The only correct answer is D <i>A is not correct because bacterial cells do not have a cellulose cell wall</i> <i>B is not correct because plant cells do not have pili</i> <i>C is not correct because plant cells do not have a capsule</i>	(1)

Question Number	Answer	Additional guidance	Mark
3(c)(ii)	An explanation that includes two of the following points: • optimum temperature for enzyme reactions (to occur) / {suitable/ best} temperature for {optimum / fastest / increased rate of} enzyme reactions (to occur) (1) • optimum pH for enzyme reactions (to occur) / suitable pH for {optimum / fastest / increased rate of} enzyme reactions (1) • (some bacteria require) oxygen for (aerobic) respiration (1)	ACCEPT metabolic reaction for enzyme reaction ACCEPT named enzyme reaction e.g. respiration ACCEPT suitable temperature so enzymes do not denature ACCEPT {suitable / optimum} temperature for enzymes to work best at} ACCEPT suitable pH so enzymes do not denature ACCEPT {suitable / optimum} pH for enzymes to work best at} ACCEPT (some bacteria require) no oxygen for anaerobic respiration	(2)